Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Real-time flow cytometry coupled with microlens array single-cell imaging**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (X), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Advanced Bio		Advanced Bio	
	Medium Category	Small Category	Medium Category	Small Category
	Bio-processes / Bio-Equipment	Bio-Analytical Instruments	Medical Bio	Cell and Tissue Therapeutics
Concept of Proposed Technology	This technology integrates real-time flow cytometry with microlens-array-enhanced single-cell imaging, enabling simultaneous high-speed fluorescence detection and sorting, and localized optical magnification. It provides rapid, rich, and spatially resolved single-cell information for advanced analytical workflows.			
Rationale for Joint R&D Project	As an innovation center, we focus on developing advanced enabling technologies that integrate seamlessly into partners' instruments and workflows. Joining this R&D project allows us to co-create a modular platform that strengthens Korean SME's products while accelerating technology transfer and real-world adoption.			
Key Words	1. Real-time flow cytometry 2. Advanced single-cell sorting 3. Microlens array imaging 4. Inline bioprocess analytics			

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 4	End of the Project	Level 6

R&D Objectives	The project aims to develop an integrated technology that combines real-time flow cytometry with microlens-array-enhanced single-cell imaging, enabling high-speed optical characterization with improved spatial resolution. As an innovation center, we focus on creating modular, robust components that can be seamlessly transferred and integrated into commercial instruments and analytical workflows.
	
	This will be achieved by:
	• Adapting the real-time fluorescence detection and analysis module to the SME's need, including
	  ○ FPGA-based signal processing
	  ○ multi-channel acquisition
	  ○ automated gating for accurate identification of single fluorescent events at high throughput.
	• Implementing real-time sorting capabilities, enabling
	  ○ selective routing of droplets or cells based on fluorescence-derived criteria,
	  ○ peak structure,
	  ○ multi-channel gating
	  ○ compact, embeddable sorting solution for partner systems.
	• Engineering and validating microlens array (MLA) properties, covering
	  ○ lens design
	  ○ number
	  ○ layout
	  ○ optical simulation
	  ○ fabrication
	  ○ characterization
	  ○ adaptation to target hardware (microfluidic chip, glass, COC)
	• Demonstrating application-relevant use cases,
	  ○ improved single-cell classification
	  ○ image-based discrimination
	  ○ enhanced data richness in representative biological workflows			

R&D Target Specifications	• Real-time fluorescence analysis
	  ○ ≥1 MHz sampling rate
	  ○ ≥ 4 fluorescence channel detection
	  ○ multi-parameter gating
	  ○ FPGA-based peak extraction
	• Sorting capability			

	○ selective routing of droplets or cells in 2 separate classes ○ adaptable delays ○ user-defined fluorescence criteria • Microlens array ○ magnification between 3–10× ○ consistent optical quality (sag, ROC, focal shift) ○ reliable fabrication on glass substrates. • Verstatile target hardware, such as microfluidic chips and their materials.
R&D Plan	Concept selection • Agile development of concepts while defining the requirements • Evaluate the hardware required for the fluorescence signal, including laser lines, sensitivity • Compare microlens arrays design, substrates and fabrication methods (UV-Nil, hot embossing) • Perform tests to validate feasibility Expected outcome: selected concept with documented requirements and validation results Prototype development and characterization • Manufacture of the hardware for fluorescence excitation and signal collection • Fabricate microlens arrays on desired substrate and characterize main optical properties • Validate real-time fluorescence analysis • Demonstrate functional sorting Expected outcome: fully functional and characterized prototype meeting target specifications. Prototype deployment • Support installation, calibration and training of users • Final reporting including performance metrics Expected outcome: prototype functional and successfully deployed at the partner.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	CSEM will lead the development, integration, and validation of the enabling technologies combining real-time fluorescence cytometry with microlens-array-enhanced single-cell imaging. Our role focuses on creating robust, modular technological components that can be directly transferred into partners' analytical instruments and workflows.
	
	• Scientific and technical development of the real-time fluorescence analysis module, including
	  ○ FPGA-based signal processing
	  ○ multi-channel detection
	  ○ automated gating strategies
	  ○ generation of sorting triggers for high-throughput droplet or cell routing.
	• Design, fabrication, and characterization of microlens arrays, covering
	  ○ optical simulation
	  ○ UV-NIL or embossing-based microfabrication
	  ○ validation of magnification performance within standard labware and microfluidic chip architectures.
	• Development of modular hardware and software interfaces, enabling
	  ○ straightforward adoption by industrial partners
	  ○ compatibility with microfluidic screening systems.
	• Structured technology transfer, including
	  ○ Documentation
	  ○ Training
	  ○ knowledge exchange to support partners in embedding the developed modules into their own products or R&D workflows.
	• Execution of
	  ○ application-driven demonstration studies
	  ○ validating system performance under relevant biological use cases
	  ○ delivering high-quality datasets to support integration and decision-making.
	.

Expected R&R of Korean SME Partner	The Korean industry partner will act as an application-driven development collaborator, providing essential industrial requirements, practical constraints, and use-case scenarios to ensure that the integrated fluorescence-sorting and microlens-array imaging technology is aligned with market needs and ready for downstream adoption.
	
	• Provision of application cases and operational requirements, including
	  ○ sample types
	  ○ workflow constraints
	  ○ desired throughput
	  ○ imaging needs
	  ○ performance criteria relevant to their diagnostic, analytical, or manufacturing environment.
	• Supply components required for developing and validating the integrated flow-cytometry and MLA-enhanced imaging modules under realistic conditions, such as:
	  ○ representative materials

	○ biological samples ○ device components • Execution of complementary in-house testing, performing ○ comparative measurements or routine characterization to ○ support validation of system performance, robustness, and usability in an industrial setting. • Active participation in iterative development, providing continuous feedback on ○ prototype functionality ○ data output ○ integration needs ○ user-interface expectations to guide refinement of both hardware and software interfaces. • Collaboration on deployment and standardization, helping ○ adapt the developed modules to their internal systems ○ documenting implementation challenges ○ contributing to the definition of SOPs, integration guidelines, and quality requirements. • Support for exploitation and technology transfer, including ○ evaluation of potential commercialization pathways, ○ joint dissemination activities, ○ engagement in optional IP generation or co-development of product-specific variants.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	• A fully validated prototype combining
	  ○ real-time fluorescence analysis
	  ○ sorting capability
	  ○ microlens-array-enhanced single-cell imaging,
	• Transferable hardware and software modules enabling Korean industry partners to integrate high-speed optical readout and MLA-based imaging into their own analytical devices, screening platforms, or diagnostic workflows.
	• Accelerated development cycles for the Korean partner, enabled by earlier access to high-quality, quantitative single-cell data and improved characterization of biological samples, materials, or product candidates.
	• Strengthened technological capabilities within the Korean ecosystem, including enhanced competence in real-time optical analysis, micro-optical components, and integrated imaging workflows, fostering local innovation and reducing reliance on imported technologies.
	• Opportunities for future co-development or commercialization, including potential joint IP, adaptation of the modules to specific industrial products, and expansion into new markets requiring compact, high-performance single-cell analysis solutions.

6. References

(1)

(2)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization